



RiskPulse

An AI/NLP risk engine that turns financial news and social chatter into structured, machine-readable risk signals — and acts on them.

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VIT-AP University · Individual submission

Risk lives in words. Systems can't read them.

VOLUME

Thousands of headlines and posts an hour — far beyond what an analyst desk can read.

STRUCTURE

Raw text can't feed a model, a limit system or a portfolio revaluation.

LATENCY

Risk teams learn about the shock after the P&L does — at the overnight run.

MY APPROACH

01 Ingest

News CSV, social JSON and any live RSS feed → one Document format.

02 Understand

Link companies, score sentiment, classify the event, predict impact.

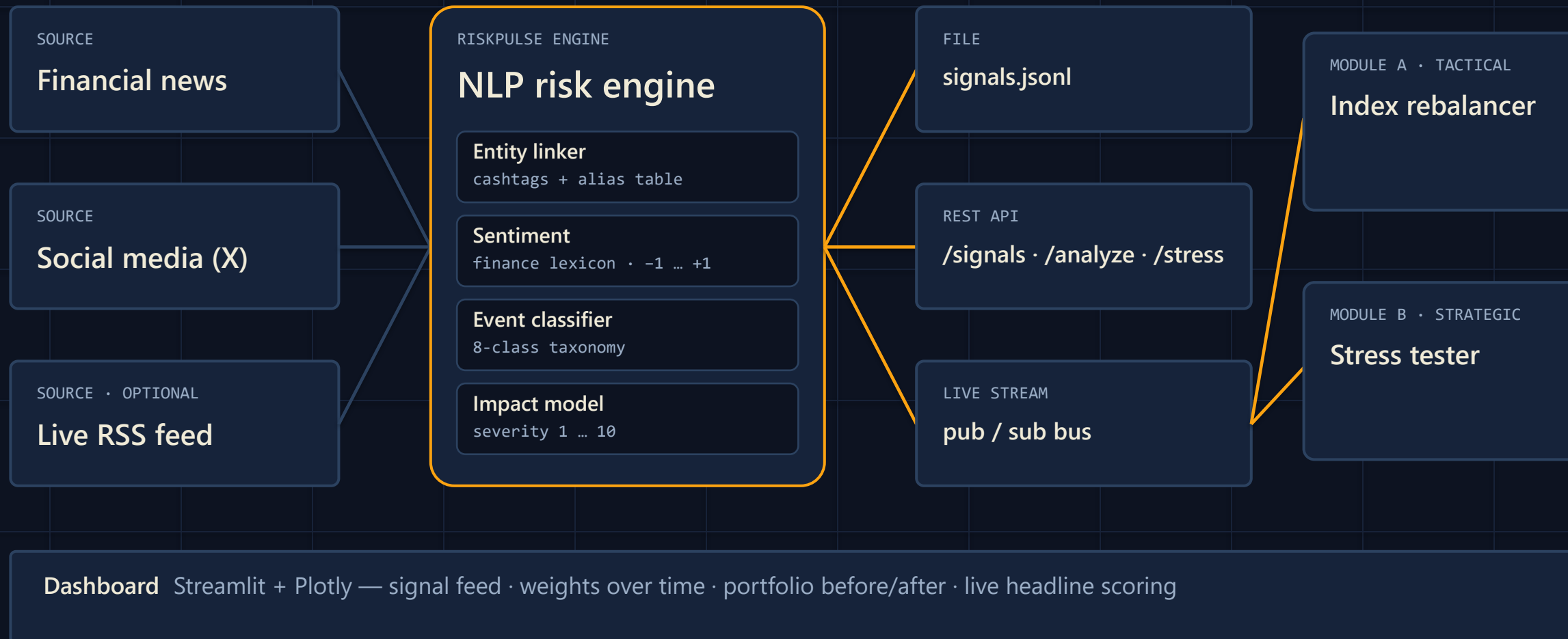
03 Publish

Signals to a JSONL file, a REST API and a live pub/sub stream.

04 Act

Module A tilts an index · Module B stress-tests a banking book.

One engine. Two consumers. Three ways out.



One headline in. One signal out.

INCOMING · NEWS WIRE

Rating agency downgrades Goldman Sachs after surprise trading losses; contagion fears hit bank stocks

ENTITY

GS

alias → ticker

SENTIMENT

−0.84

lexicon, negation-aware

EVENT

Credit Event

confidence 100%

IMPACT

8.6 / 10

triggers a stress test

Explainable lexicon, not a black box

~250 finance terms + phrases. 'Crushed it' is positive, 'higher for longer' negative. Every score ships with the words behind it — auditable for model risk.

Impact is a model, not a guess

Event base severity + tone (negative $\times 1.25$) + magnitude cues (7%, \$16bn, 25bp) + shock words + breadth + source credibility by follower reach.

Modules built like desk tools

A: $w \propto \text{base} \cdot e^{(k \cdot s)}$ with floors, a 15% cap and 10% turnover limit. B: duration / spread / delta / FX / PD $\times$ LGD revaluation; scenario by event, scaled by impact.

One sample week, end to end.

60

DOCUMENTS

68

RISK SIGNALS

46

REBALANCES

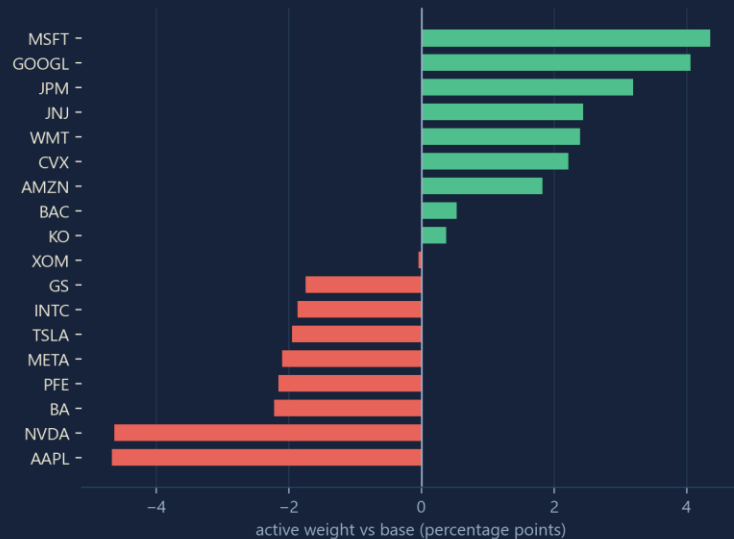
10

STRESS TESTS

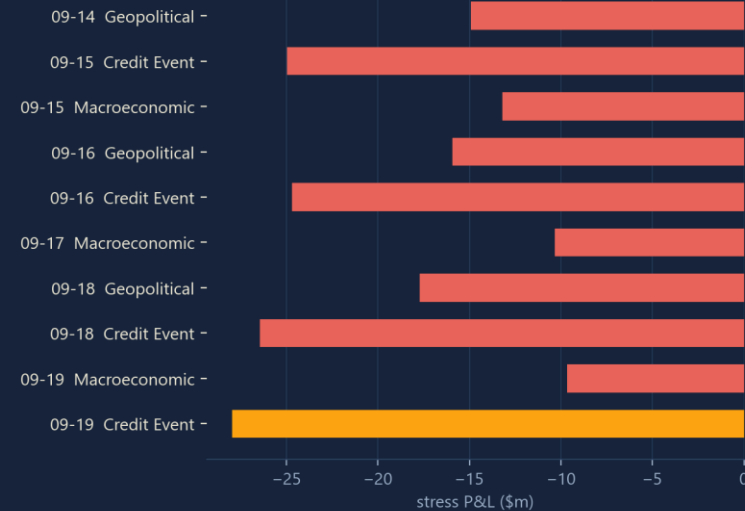
WORST CASE · SOVEREIGN DOWNGRADE · IMPACT 9.4

—\$28.0M

-9.9% of a \$283m book, valued the moment it landed



Module A — AAPL, NVDA cut ~4.6pp; MSFT, GOOGL, JPM up



Module B — 10 auto-triggered stress tests (amber = worst)

News and social agree on sentiment direction in 29 of 30 stories — the miss (OPEC+ cut) is genuinely mixed news.

Why it matters to a bank or an index provider.

EARLY WARNING

Seconds, not overnight

A credit event in the news revalues the book as it lands — instead of at the next daily risk run.

EXPLAINABILITY

Every number has a reason

Scores trace back to the words that produced them, so model validation and regulators can audit them.

ONE SIGNAL, MANY USES

Build once, consume everywhere

The same feed drives trading tilts, scenario analysis and — next — limit monitoring and alerts.

ANALYST LEVERAGE

One ranked feed

A desk watching 18 names and 60 stories sees what matters first, instead of dozens of tabs.

Honest limits, clear next steps.

ASSUMPTIONS & GAPS

- Lexicon tuned on a small synthetic sample
- First-order (linear) revaluation — no convexity or optionality
- Scenario shocks hand-calibrated, not fitted to history
- Entity linking limited to an alias table

NEXT STEPS

- Fine-tune FinBERT on labelled headlines; ensemble with the lexicon
- Calibrate impact against realised abnormal returns
- Historical scenario library — 2008, 2020, 2022 rates shock
- Streaming ingestion (Kafka) with alerting; backtest Module A vs equal weight